

EEE/ECE 308L (Electronic Circuits II Laboratory)

Department of EEE, Brac University

Marks: 30

Suppose you want to see yourself as an audio engineer who has obtained a degree in acoustic engineering. Now, you dream of getting a job in a renowned speaker manufacturing company and one day you come across a job circular posted by Sony Corporation. When you approach them with your résumé, they select you for a test where you are supposed to solve the following problem.

You are tasked with the design of a tweeter speaker. The speaker is supposed to reject all sounds having a frequency below a certain threshold which is given below.

$$f_{thres} = [10 + G] \text{ kHz}$$

Here, G is your group number. You need to design a 3rd-order active filter in order to meet the requirements.

Your design should be following the standard design producers give below-

Firstly, you will design based on theories and concepts to decide the circuit components and configuration. **Secondly**, you should use simulation to verify the design. **Finally**, the hardware implementation of the design should be developed and tested for justification.

Refer to the guidelines attached in the next page for your submission.

Following rubrics are going to be used to evaluate your task-

Performance Criteria	Performance Measure			
	Unsatisfactory (1)	Developing (2)	Satisfactory (3)	Exemplary (4)
1. Design and set up experiments. Mark: 5	Not able to design or setup the experiment at all	Minimal mention of design and setup of experiment	Can Mentions design and setup appropriately	Mentions design and setup elegantly
2. Conduct experiments and perform measurements. Mark: 8	Does not mention of conduct of experiments and perform measurements	Minimal mention of conduct of experiments and perform measurements	Can conduct experiments and perform measurements appropriately	Experiment is conducted elegantly

3. Detect the experimental faults and troubleshoot them Mark: 7	Does not know how to detect experimental faults and troubleshoots	Minimal mention of detecting experimental faults and troubleshoots	Can detect experimental faults and troubleshoots appropriately	Detect experimental faults and troubleshoots elegantly
4. Analyze and Interpret data, graph, charts etc. Mark: 5	Unable to analyze and interpret data, graph, charts at all	Can analyze and interpret data, graph, charts but with poor notation, or not rigorously	Can analyze and interpret data, graph, charts rigorously in most cases	Always perform the analysis and interpretation of data, graph, charts very thoroughly
5. Synthesize the information and findings obtained through research and/or experiments to reach valid conclusions Mark: 5	Unable to synthesize the information and findings	Able to synthesize the information and findings and obtain valid conclusion in some cases.	Can synthesize the information and findings and obtain valid conclusion in most cases.	Always able to synthesize the information and findings of the research and experiments. Obtain valid conclusion with clarity

Attributes of complex engineering problem-

	Attribute	
P1	Depth of knowledge required:	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P3	Depth of analysis required:	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
P7	Interdependence:	High level problems including many component parts or sub-problems.

Guidelines:

The simulation files and a report (.pdf) should be submitted. The report must include the following sections:

1. Specified Problem statement:

[You need to identify the specification/s based on the problem statement provided]

2. Design of the circuit:

[Provide all calculations and consideration for the design of your problem]

3. Simulation result:

*[Provide the simulation **design** with **parameters considered** and the **result/s along with appropriate explanation**. Adequate analysis and interpretation of simulation graphs are expected i.e. cut-off frequency, slope of the frequency curve]*

4. Hardware implementation:

4.1. Formulating experimental setup based on the simulated design-

[You need to choose the components of your design maintaining close proximity with simulated design, mention the component values and provide a picture of your hardware setup]

4.2. Perform experiments and measurements-

[Consider adequate data (data table) for the characteristic frequency plot of the designed filter; interpret the frequency plot mentioning cut-off frequency, slope roll off and percentage error in cut-off frequency than that of simulated result; provide adequate explanation]

4.3. Troubleshoot-

[Troubleshoot the experimental setup to reduce the error percentage in cut-off frequency; provide the new data table, frequency curve along with cut-off frequency and highlight the error reduction; provide adequate explanation]

5. Summary of findings:

[Summarize the findings of software and hardware implementations and experiments to conclude and justify your design]

General Guidelines:

- Report should contain the provided cover page.
- Each group will submit only once. Multiple submissions are not acceptable. The naming convention of the files will be “OEP_Group No”. For example, if your group number is 5, then the files’ name should be “OEP_5”.
- The report can be a mixture of handwritten and computer composed parts.

- You also need to prepare a video demonstration of your project explaining the workings of the hardware and software portions. The length of the video should not exceed 5 minutes.
Attach the link to the video at the cover page of your report.
- At the end of the report mention the contributions of each group member.